

Bus Booking Analytics System

(MySQL + Python + Power BI)

1. Project Overview

The Bus Booking Analytics System is an end-to-end data analytics solution designed to analyze booking operations, customer behavior, and revenue trends. The system integrates MySQL for data storage, Python for ETL and data processing, and Power BI for visualization and business intelligence.

2. Objectives

- Design a scalable relational database for bus booking operations
 - Generate and manage large volumes of booking data (1000+ records)
 - Perform ETL (Extract, Transform, Load) using Python
 - Apply data validation and cleansing techniques
 - Build interactive dashboards for business insights
 - Enable decision-making using data-driven metrics
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3. System Architecture

Components:

1. Data Source

2. Simulated booking data (CSV files)

3. Database Layer (MySQL)

4. Stores structured relational data

5. ETL Layer (Python)

6. Data extraction from CSV
7. Data transformation & validation
8. Data loading into MySQL

9. Visualization Layer (Power BI)

10. Dashboard creation

- 11. KPI monitoring
 - 12. Trend analysis
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4. Database Design (High-Level)

Key Tables:

- **Customers**

- Customer_ID (PK)
- Name
- Email
- Phone

- **Buses**

- Bus_ID (PK)
- Bus_Number
- Bus_Type (AC/Non-AC/Sleeper)
- Capacity

- **Routes**

- Route_ID (PK)
- Source
- Destination
- Distance

- **Bookings**

- Booking_ID (PK)
 - Customer_ID (FK)
 - Bus_ID (FK)
 - Route_ID (FK)
 - Booking_Date
 - Travel_Date
 - Seat_Number
 - Fare_Amount
 - Booking_Status
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5. Data Generation Strategy

- Generate synthetic data using Python libraries (e.g., Faker, random)

- Ensure realistic patterns:
 - Peak bookings on weekends
 - Popular routes with higher demand
 - Different fare ranges based on distance and bus type
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6. ETL Process (High-Level)

Extract:

- Load CSV files into Python (Pandas)

Transform:

- Clean data
- Apply validation rules
- Format columns
- Remove inconsistencies

Load:

- Insert cleaned data into MySQL tables
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7. Data Validation Techniques

1. Duplicate Removal

- Remove duplicate booking records based on Booking_ID

2. Handling Null Values

- Replace missing values with defaults or remove invalid rows
- Example:
- Missing fare → remove record
- Missing phone → mark as "Unknown"

3. Data Standardization

- Date format → YYYY-MM-DD
- Text fields → consistent casing
- Phone numbers → uniform format

4. Data Consistency Checks

- Ensure foreign key relationships are valid
- Validate:
- $\text{Travel_Date} \geq \text{Booking_Date}$

- Fare_Amount > 0
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8. Power BI Dashboard (High-Level)

Key Visualizations:

-  Total Revenue (KPI)
-  Booking Trends Over Time
-  Customer Distribution
-  Route Performance
-  Seat Occupancy Rate
-  Top Routes by Revenue

Filters:

- Date Range
 - Route
 - Bus Type
 - Booking Status
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9. Key Business Metrics

- Total Bookings
 - Total Revenue
 - Average Fare per Booking
 - Customer Retention Rate
 - Occupancy Rate
 - Most Popular Routes
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10. Expected Insights

- Identify peak booking periods
 - Discover high-revenue routes
 - Analyze customer behavior patterns
 - Optimize pricing strategy
 - Improve resource allocation (buses, routes)
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11. Tools & Technologies

Layer	Tool
Database	MySQL
ETL	Python
Visualization	Power BI

12. Conclusion

This project demonstrates a complete data pipeline from raw data generation to business insights. It highlights real-world data challenges and showcases how analytics can drive operational efficiency and strategic decisions in the transportation domain.